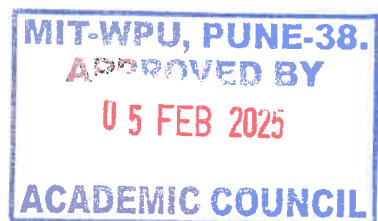




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Course Code	DCA20010				
Course Category	PF				
Course Title	Database Management System				
Teaching Scheme	L	T	Laboratory/Practical	Project	Total
Weekly Load Hrs.	3	-	2	-	5
Credits	3	-	1	-	4
Assessment Schema Code	TL3				

Pre-requisites:

Basic Mathematics, Logical Reasoning.

Course Objectives:

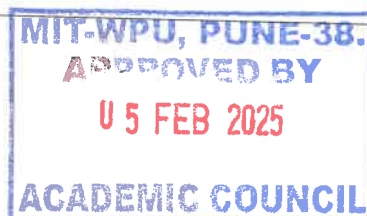
1. To introduce data processing using computers.
2. To explain data models used for database design
3. To explain the normalization of database schema
4. To understand creations, manipulation and querying of data in databases

Course Outcomes:

On completion of the course, student will be able to—

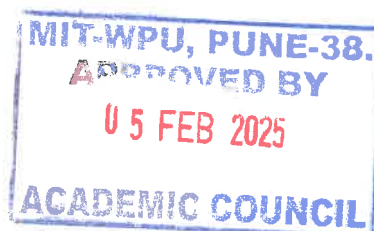
1. Understand the fundamental concepts of database management systems, including their structure, components, and advantages over traditional file-processing systems.
2. Analyze and design databases using the Entity-Relationship (ER) model, mapping real-world scenarios into ER diagrams.
3. Apply relational database concepts, including keys and schema design, to effectively model data and normalize databases to 3NF.
4. Evaluate the principles of relational database design, including functional dependencies, normalization, and decomposition to ensure efficient and reliable databases. Utilize SQL to create, query, and manipulate databases, leveraging advanced techniques such as joins, nested queries, and aggregate functions.

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Course Content		
Sr. No.	Topics Details	No of Sessions (Hours)
1	Unit 1: Introduction to Database Management Systems 1.6 Drawbacks of using files to store data, 1.7 purpose of database systems, and the definition of DBMS 1.8 Comparison of file processing systems and DBMS, 1.9 limitations of file processing systems, 1.10 advantages and disadvantages of DBMS, 1.11 users of DBMS, and overall system structure	6
2	Unit 2: Entity-Relationship (ER) Model 2.1 Entities, 2.2.entity sets, 2.3. relationships, 2.4.relationship sets, and attributes. 2.5. Mapping cardinalities and ER diagrams. 2.6.Case studies demonstrating the use of the ER model in database design.	6
3	Unit 3: Relational Model and Database Design 3.5 Structure of relational databases and terms: relation, tuple, attribute, and cardinality. 3.6 Key concepts including super key, candidate key, primary key, and foreign key. 3.7 Conversion of ER diagrams to relational models and relational schemas to 3NF. 3.8 Functional dependencies, closure of functional dependencies (F+), closure of an attribute set, and the algorithm to derive a primary key. 3.9 Case studies on relational models.	15

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4	Unit 4: Normalization and Relational Database Design 4.1 Pitfalls in relational databases, the concept of decomposition, and desirable properties. 4.2 Overview of normalization, including 1NF, 2NF, and 3NF, with algorithms and examples for implementation. 4.3 Case studies illustrating database normalization techniques.	10
5	Unit 5: Structured Query Language (SQL) 1.1 Introduction to SQL, its history, and basic structure. Classification of SQL statements, 2.1 DDL commands (CREATE, DROP, ALTER), and DML commands (INSERT, UPDATE, DELETE, SELECT). 3.1 Data types, constraints (Primary Key, NOT NULL, UNIQUE, CHECK, Foreign Key, Default), and SQL operators (IN, NOT IN, BETWEEN, LIKE). 4.1 SQL built-in functions, including string, numeric, date, and time functions. 5.1 Advanced SQL queries covering aggregate functions, set operations, ORDER BY, GROUP BY, HAVING clauses, joins (inner and outer), and nested queries with case studies.	8

Reference Books:

- Database System Concepts, Henry korth and A. Silberschatz
- An Introduction to Database System, Bipin Desai
- File Structure by Michael, J. Folk, Greg, Riccardi
- Teach Yourself SQL in 14 days, Jeff Parkins and Bryan Morgan

Web Resources

Weblinks:

- <https://www.geeksforgeeks.org/dbms/>
- <https://www.w3schools.com/sql/default.asp>

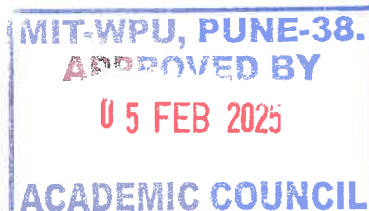
MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc

Pedagogy:

- Participative Learning,
- Discussion
- Demonstrations
- Practical
- Assignment
- Tutorials
- Lab assignments

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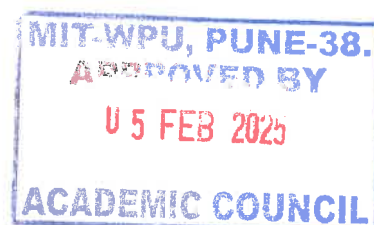




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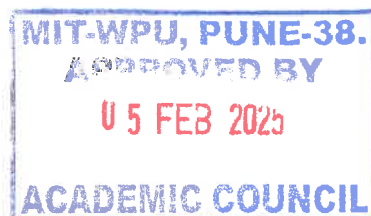
Course Structure					
Course Code	BSC10050				
Course Category	PM				
Course Title	Microprocessor				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3		2		5
Credits	3		1		4
Assessment Schema Code	TL3				
Prerequisites: Basic of digital electronics, Basic terms related to Microprocessor					
Course Objectives: 1. To understand the design of the architecture of 8086 Microprocessor 2. To understand the programming model of 8086 3. To understand how to write an assembly language program. 4. To understand the Advanced Microprocessor					
Course Outcomes: On completion of the course, student will be able to— 1. Understand the block diagram of 8086 Microprocessor 2. Understand the software model of 8086 3. Write an assembly language program 4. Understand the advanced Microprocessor					

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Course Content		
Sr. No.	Topics Details	No of Sessions (Hours)
1	Unit 1: 8086 Architecture 1.1 Evolution of Microprocessors and types, 1.2 8086 Microprocessor, Salient features, Pin descriptions, 1.3 Architecture of 8086 - Functional Block diagram, Register organization, 1.4 Concepts of pipelining (5 stage), Memory segmentation, Physical memory addresses generation., 1.5 Minimum mode and Maximum Mode of 8086	10
2	Unit 2: Instruction Set of 8086 Microprocessor 1.1 Addressing modes, Data transfer instructions, Arithmetic Instructions, Logical Instructions, 1.2 Bit manipulation instructions, String Operation Instructions, 1.3 Program control transfer or branching Instructions, 1.4 Process control Instructions, examples based on Instruction Set	15
3	Unit 3: 8086 Assembly Language Programming 3.1 Introduction to Assembly Programming and Assembler Directive 3.2 Assembly program on Addition, Subtraction, Multiplication and Division 3.3 Sum of Series Smallest and Largest numbers from array 3.4 Sorting numbers in Ascending and Descending order Block transfer 3.5 String Operations - Length, Reverse, Compare, Concatenation, Copy	15
4	Unit 4: Advanced Processor 4.4 Introduction to Pentium Processor, 4.5 Feature of Pentium Pro Processor Architecture of Pentium Pro Processor, 4.6 Software Model of Pentium Pro , 4.7 Pipelining of Pentium Processor , 4.8 Classification of Advanced Processors	10

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Reference Books:

1. Microprocessor & interfacing (programming & hardware) Revised Second Edition By Douglas V. Hall Tata McGraw Hill
2. Microprocessor Architecture By B.Ram
3. Microprocessor Architecture, Interfacing & Programming By Mohammad Raffiqzaman

Supplementary Reading:

- 1) The Intel Microprocessors : Barry B. Brey- Pearson Education Asia
- 2) The Pentium Microprocessor : James Antonakos

Weblinks:

1. www.intel.com
2. www.pcguide.com/ref/CPU
3. www.CPU-World.com/Arch/

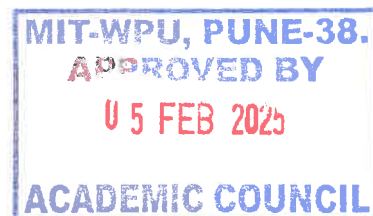
MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc

Pedagogy:

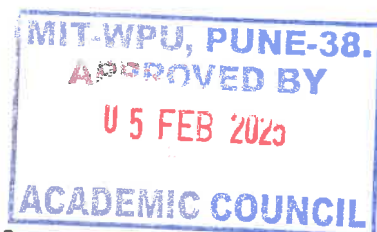
- o Participative learning
- o Discussions
- o Demonstrations
- o Demonstrations
- o Problem solving

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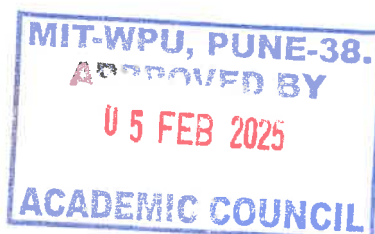


Course Structure					
Course Code	DCA20020				
Course Category	PF				
Course Title	Data Structure using C				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3		2		5
Credits	3		1		4
Assessment Code	TL3				
Prerequisites: 1. Basic knowledge of computer programming terminologies 2. Knowledge of C Programming language 3. Introduction to problem solving					
Course Objectives: 1. To introduce the fundamental concept of data structures 2. To emphasize the importance of data structures in developing and implementing efficient algorithms.					
Course Outcomes: After completing this course, a student will be able to: 1. Understand the concepts of data structures, including primitive and non-primitive types, linear and non-linear structures, and analyze algorithms using Big O notation. 2. Develop skills in creating, traversing, and manipulating different types of linked lists, including singly, doubly, and circular linked lists. 3. Demonstrate the implementation and applications of stacks and queues, including operations like insertion, deletion, and evaluation of expressions using stacks. 4. Understand and implement binary trees, binary search trees (BST), AVL trees, and heaps, and perform tree traversals (Inorder, Preorder, Postorder). 5. Apply graph traversal algorithms (DFS, BFS), implement shortest path algorithms (Dijkstra's, Bellman-Ford), and develop efficient searching and sorting algorithms.					



Course Content		
Sr. No.	Topics Details	No of Sessions (Hours)
1	Unit 1: Introduction to Data Structures 1.1 Self-Referential Structure, 1.2 Data Structures: Overview and classification, 1.3 Primitive and Non-Primitive Data Structures, 1.4 Linear and Non-linear Structures, Algorithm Analysis, Big O Notation	7
2	Unit 2: Linked Lists 2.1 Representation of Linked Lists: Static and dynamic representation 2.2 Singly Linked List: Creation, Insertion: Begin, Middle, End, Printing and Deleting: Begin, Middle, End, Traversing the list, 2.3 Doubly Linked List: Creation, Deletion, Circular Singly Linked List: Creation, Deletion	11
3	Unit 3: Stack and Queue 3.1 Stack: Static and Dynamic Representation, Operations, Applications: Postfix Expression Evaluation, Infix to Postfix Conversion, String Reversal 3.2 Queue: Static and Dynamic Representation, Operations, Priority Queue and Circular Queue Implementation	8
4	Unit 4: Trees 4.1 Introduction to Trees: Definition and Terminology, Tree Representation, 4.2 Binary Tree: Representation and Traversal (Inorder, Preorder, Postorder, 4.3 Binary Search Tree (BST): Implementation, Heap, AVL/Height-Balanced Tree Classification of Advanced Processors	9

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5	Unit 5: Graphs, Searching, and Sorting 5.1 Graphs: Introduction and Terminology, Representation of Graphs: Adjacency Matrix and Adjacency List, Graph Traversals: Depth First Search (DFS) and Breadth First Search (BFS), Shortest Path Algorithms: Dijkstra's and Bellman-Ford Algorithms 5.2 Searching: Types of Searching Algorithms: Linear Search, Binary Search 5.3 Sorting: Types of Sorting Algorithms: Bubble Sort, Insertion Sort, Merge Sort, Selection	10
Reference Books: 1. Fundamentals of Data structures, . Horowitz and S. Sahani 2. Introduction to Data Structures in C, Ashok N. Kamthane 3. Data Structure Using C, Radhakrishnan and Shrivastav 4. Data Structure Using C, Bandopadhyay & Dey(Pearson) Text Books: 1. "Introduction to Algorithms", Thomas H. Charles E. Leiserson, Ronald L. Rivest and Clifford Stein.Cormen 3rd Edition (The MIT Press) 3rd Edition 2. Data Structures and Algorithms Made Easy,Narsimha Karumanchi 3. Algorithms,Robert Sedgewick and Kevin Wayne. Websites: • https://www.programiz.com/dsa • https://www.w3schools.in/data-structures/intro Pedagogy: • Participative Learning, • Discussion • Demonstrations • Practical • Assignment		

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Lab on Data structures

Sr.No	Practical to be conducted on	Lab Hrs
1.	1.1.1 Introduction to Data Structures • Objective: Learn and implement the basics of data structures. • Tasks: ○ Understand and create primitive and non-primitive data structures (arrays, lists, etc.). ○ Analyze an algorithm using Big O notation.	2
2.	1.1.2 Linked List Operations • Objective: Implement operations on singly and doubly linked lists. • Tasks: ○ Create a singly linked list. ○ Perform insertion at the beginning, middle, and end. ○ Perform deletion at the beginning, middle, and end. ○ Implement and traverse a doubly linked list. ○ Create a circular singly linked list.	2
3	1.1.3 Stack Implementation • Objective: Implement stack data structure using both static and dynamic representations. • Tasks: ○ Implement stack operations (push, pop, peek). ○ Evaluate postfix expressions using stack. ○ Convert infix expressions to postfix form. ○ Reverse a string using a stack 1.1.4 Queue Implementation	2

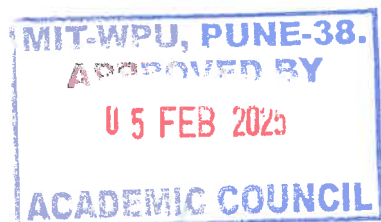
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ACADEMIC COUNCIL

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	• Objective: Implement queue data structure using both static and dynamic representations. • Tasks: ○ Implement basic queue operations (enqueue, dequeue). ○ Implement a circular queue and priority queue.	
4	1.1.5 Binary Tree Traversals • Objective: Implement binary tree data structure and perform tree traversal techniques. • Tasks: ○ Create a binary tree. ○ Perform Inorder, Preorder, and Postorder traversals. ○ Implement a Binary Search Tree (BST) and insert nodes. 1.1.6 AVL Tree Implementation • Objective: Implement an AVL tree (self-balancing binary search tree). • Tasks: ○ Create an AVL tree. ○ Implement insertion and deletion in the AVL tree with rotations for balancing.	3

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5	1.1.7 Graph Representation and Traversals • Objective: Understand and implement graph data structures and traversal algorithms. • Tasks: ○ Represent graphs using adjacency matrix and adjacency list. ○ Implement Depth-First Search (DFS) and Breadth-First Search (BFS) graph traversal. 1.1.8 Shortest Path Algorithms • Objective: Implement shortest path algorithms in a graph. • Tasks: ○ Implement Dijkstra's algorithm. ○ Implement Bellman-Ford algorithm to find the shortest path. 1.1.9 Searching Algorithms • Objective: Implement different searching algorithms. • Tasks: ○ Implement linear search. ○ Implement binary search for sorted arrays. 1.1.10 Sorting Algorithms • Objective: Implement various sorting algorithms. • Tasks: ○ Implement Bubble sort, Insertion sort, and Selection sort. ○ Implement Merge sort and Quick sort. 1.1.11	6
	Total Hrs	15

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